

penalizing deviations

Unbalanced OT: Formulation

- $\mathbf{a} \in \mathbb{R}_+^n$, $\mathbf{b} \in \mathbb{R}_+^m$, not necessarily adding up to 1, can have $\sum_i \mathbf{a}_i \neq \sum_j \mathbf{b}_j$

- Relax marginal constraints by **penalizing deviations**:

$$\min_{\pi \geq 0} \langle C, \pi \rangle + \tau_1 D_\psi(\pi \mathbf{1}_m \mid \mathbf{a}) + \tau_2 D_\psi(P^\top \mathbf{1}_n \mid \mathbf{b})$$

where D_ψ is a divergence (often KL).

- Smaller $\tau_1 \rightarrow$ the more the marginal $\tilde{\mathbf{a}} \triangleq \pi \mathbf{1}_m$ is allowed to deviate from $\mathbf{a}$
- Recovers:
 - Classic (balanced) OT when $\tau_i \rightarrow \infty$ (if $\sum_i \mathbf{a}_i = \sum_j \mathbf{b}_j$)
 - Hellinger distance for $D_\psi = \text{KL}$, $c(x, y) = -\log \cos(\min(d(x, y)/\kappa, \pi/2))$, $\tau_i \rightarrow 0$

Unbalanced OT: Intuition